

FLOW METERING Open Channel Flowmeters



Hach/GLI sc200 Open Channel Flow Controllers

- Two 4-20 mA outputs and four SPDT relays
- Measure a second input with optional flow module

Hach's sc200 ultrasonic flow controller replaces and improves upon the GLI U53 controller series. It gives reliable, highly accurate flow and depth measurement using ultrasonic technology. The intuitive multiple-language menu-driven system guides you through operation sequences.

The controller includes a large range of easy-to-use preprogrammed flume and weir calculations, or create your own via user-entered tables. View flow rate, flow total, depth and volume directly on the 3" x 2.5" backlit high-resolution display. These controllers also datalog up to 20,000 data points. The optional SD card and SD card reader allow simple download and transfer of event history, logged data, and controller settings to a computer.

Controllers feature four SPDT relays to trigger alarms and timers, or for on/off or proportional control of pumps and other devices. Assign the two 4-20 mA outputs to any measurement, including proportional or PID control for chart recorders, chemical metering pumps, data loggers or SCADA systems. The optional 4-20 mA output module adds an extra four 4-20 mA outputs.

Order a secondary input module to measure a second channel of flow. The 1/2 DIN NEMA 4X (IP65) enclosure can be surface-, panel- or pole-mounted with the included mounting kit.

Note: Order both the U53 controller and the U53 sensor with the preferred cable length for a complete system. Cable is integral to the sensor and cannot be field-spliced.

Controller specifications

Accuracy:	0.5% of span
Display:	backlit graphic dot matrix LCD, 240 x 160 pixels
Analog output:	dual 4-20 mA isolated current (add 4 more with optional 4-20 mA module)
Control relays:	four SPDT relays, rated for 5A at 230 VAC, or 24 VDC
Power:	115/230 VAC, 50/60 Hz, or 24 VDC (depends on model)
Operating conditions:	4 to 140°F (-20 to 60°C), 0 to 95% RH non-condensing
Enclosure:	1/2 DIN (7"W x 10"H x 4 1/4"D) NEMA 4X housing with hardware for surface, panel, or pipe mounting

Sensor specifications

Range:	0.82 to 19.68 feet
Resolution:	±0.04"
Temp. compensation:	integral temperature sensor
Construction:	NEMA plastic body

DESCRIPTION	STOCK #	EACH	
Flow Controller, Single Input, 115/230 VDC	38053	\$	UL
Flow Controller, Single Input, 24 VDC	29182		
U53 Ultrasonic Sensor with 10' Cable	40273		
U53 Ultrasonic Sensor with 30' Cable	40274		SP
U53 Ultrasonic Sensor with 100' Cable	40275		

OPTIONAL ACCESSORIES

Flow Secondary Input Module	38056	\$
4-20 mA Module (Provides 4 More Outputs)	204661	
Secure Digital (SD) Memory Card, 4GB	67224	
SD Card Reader	67225	
Weather Shield for Controller, Metal	38058	
Sun Screen for Controller Display, Plastic	38059	
Power Cord with Strain Relief, 125 VAC, 6'	38060	

Endress+Hauser

People for Process Automation



Endress+Hauser FMU 90 Ultrasonic Open Channel Flowmeters



- Preprogrammed with most flume and weir configurations
- 4-20 mA and pulse outputs for pacing samplers, feed pumps, etc.
- Programmable relay for alarm and control

The Endress+Hauser FMU 90 open channel flowmeter offers an outstanding combination of features and quality. Designed for simple installation and maintenance, the transmitter offers preprogrammed flume and weir configurations, a large LCD display, 4-20 mA HART output, relay control outputs, totalizer driver, sampler activation and a standardized easy-to-use operator interface. A fuzzy logic implementation manager in the transmitter smoothly combines many sophisticated signal processes that automatically adapt to changing conditions, making this one of the simplest open channel flowmeters to program.

The rugged E+H sensor has no sensitive on-board electronics to fail, and offers high material buildup immunity and corrosion resistance. The standard cable length of 16 feet may be extended up to a total of 1,000 feet. An intrinsically safe version is available.



Sensor 11672

Flowmeter specifications

Resolution:	0.08" or 0.17% of measured distance
Display:	illuminated
Output:	4-20 mA analog signal (HART)
Relays:	1 SPDT, 4A @ 250 VAC (6 relays optional)
Power:	90 to 253 VAC, 50/60 Hz
Operating conditions:	-40 to 140°F
Enclosure:	NEMA 4X reinforced plastic
Dimensions:	8.46"W x 7.09"H x 5.91"D

Sensor specifications

Model:	FDU 91
Max range in liquids:	33 ft
Blocking distance:	12 in
Beam angle:	9°
Operating frequency:	43 kHz
Mounting:	1" NPT(M), PVDF
Housing:	PVDF
Protection:	NEMA 6P/IP68
Cable length:	16 ft
Dimensions:	4.33"H x 2.76"Dia
Weight:	2.4 lbs

Note: Both a flowmeter and sensor are required for a complete system.

DESCRIPTION	STOCK #	EACH
FMU 90 Open Channel Flowmeter	14135	\$
FDU 91 Sensor w/ 16' Cable	11672	
Pipe Mount Kit for Transmitter (For 1 to 2" Tubes)	14136	